

Raavi Suhas

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LinkedIn



GitHub

Portfolio

Devpost

Professional Summary

Computer Science undergraduate specializing in Artificial Intelligence with hands-on experience in **Python, C++, Machine Learning, Generative AI, LLMs, and Agentic AI**. Experienced in building autonomous AI systems, optimization applications, software projects, and RTL designs. Strong foundation in **Data Structures and Algorithms, OOP, DBMS, Operating Systems, and Computer Architecture**.

Education

Indian Institute of Information Technology Design and Manufacturing Kancheepuram	2024–2028
B.Tech in Computer Science and Engineering (AI Major)	CGPA: 8.31/10
Sri Chaitanya Junior College, Hyderabad, Telangana	2022–2024
Intermediate / Class XII	97.6%
Ratnam Global High School, Karimnagar, Telangana	2022
Class X	100%

Technical Skills

Languages: Python, C++, C, SQL

AI/ML: Machine Learning, Generative AI, LLMs, AI Agents, Agentic AI, Multi-Agent Systems, Prompt Engineering, Function Calling, Optimization

Frameworks/APIs: LangChain, LangGraph, Google Gemini API, OpenAI API, Flask, REST APIs

Libraries: NumPy, Pandas, Matplotlib, Scikit-learn, OpenCV

Cloud/Development: Google Cloud, Render, Antigravity, Git, GitHub, VS Code, Jupyter Notebook

Core CS: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Architecture

Hardware/RTL: Verilog HDL, RTL Design, ARM Architecture, Icarus Verilog, GTKWave, Kogge-Stone Adder, Barrel Shifter

Projects

OptiAgent: Autonomous Agentic Optimization System Python, Gemini, Flask – 2026

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- Independently developed and deployed an **agentic optimization platform**, building on faculty-guided research into autonomous multimodal function optimization.
- Designed an agent workflow for **landscape analysis, strategy selection, optimization, validation, monitoring, and reflection**.
- Implemented **7 optimization algorithms**: PSO, Differential Evolution, Genetic Algorithm, Simulated Annealing, Hill Climbing, CMA-ES, and Nelder-Mead.
- Integrated **Gemini Automatic Function Calling** for dynamic strategy selection, optimizer-tool interaction, independent solution validation, and structured execution.
- Implemented **convergence monitoring, stagnation detection, adaptive parameters, strategy history, and multi-stage optimization** for history-aware decision making.
- Developed a **web dashboard and Flask REST API**; used **Google Cloud, Render, and Antigravity** for development/deployment and evaluated the system on **5 benchmark functions**.

ParaPrefix-ARM: Fused Shift+Add Execute Stage Verilog HDL, ARM, RTL – 2025 GitHub

- Designed a fused ARM execute stage integrating a **logarithmic barrel shifter, Kogge-Stone parallel-prefix adder, and bypass logic** for single-cycle Shift+Add execution.
- Reduced estimated critical-path depth from **35 to 11 gate levels** and developed parameterized **32-bit Verilog RTL** modules for the datapath.
- Verified functionality using **Icarus Verilog, GTKWave, testbenches, and assertions**.

Ride Booking System C++, OOP, STL – 2026 GitHub

- Developed a C++ ride-booking application supporting **vehicle selection, driver allocation, fare calculation, OTP verification, scheduling, and feedback**.
- Applied **encapsulation, abstraction, inheritance, polymorphism, templates, STL containers, dynamic memory management, random number generation, and date/time handling**.

Certifications

- HackerRank SQL (Basic)** [Credential] **HackerRank SQL (Intermediate)** [Credential]
- HackerRank Gold Level:** Problem Solving and SQL

Achievements

- Solved **234+ programming problems** across multiple coding platforms.
- Submitted **OptiAgent** to the **All Things Agentic Hackathon**.
- Participated in coding competitions, hackathons, workshops, and technical activities.